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Changes in mangrove coverage classification criteria could impact the conservation of mangroves in Mexico

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ABSTRACT

Accurate estimates of habitat extent and rates of change are crucial inputs for the global, regional, and national assessments that guide policy-making and prioritize strategies. This can contribute to an understanding of ecosystems in the landscape for their use, management, and preservation. Mangroves are one of the types of ecosystems in which estimation discrepancies have been analyzed to determine the impacts of data quality on conservation and policy-making. We identify significant discrepancies in the extent of the last map of Mexican mangroves (i.e., 2020) produced by the Mexican Mangrove Monitoring System (MMMS). We performed a comparative assessment between the 2020 and 2015 maps by using geographical information systems to analyze the spatial extent across these years and estimate the accuracy of map changes with airborne data. We observed a spurious gain of 129,531 ha between 2015 and 2020, including 102,610 ha (79% of total changes) in the Sian Ka'an Biosphere Reserve and its surroundings. Furthermore, the mangrove definition changed, causing the MMMS to map other coastal wetlands with the presence of *Rhizophora mangle* scrubs dispersed in the landscape. The analysis of MMMS airborne data demonstrates that this significant increase is due to changes in mangrove mapping criteria and definitions. The definition and spatial delimitation of "mangrove" (i.e., mangrove community or stands forest) has implications relevant to the conservation policy of these coastal wetlands/coastal resources. MMMS discrepancies in mangrove extent could generate misleading perspectives on different sectors. A cartographic solution is to separate the other coastal wetland areas with the presence of *R. mangle* from all MMMS products and reclassify them as "other wetlands with the presence of *R. mangle*." Robust ecosystem extent data is crucial for the design and implementation of efficient land use and conservation policies.

1. Introduction

Mangroves are complex and dynamic ecosystems located in constantly fluctuating environments as coastal ecotones, and they are especially vulnerable to global change (e.g., extreme events; Jiang et al., 2016; Sharma et al., 2022). These coastal wetlands are considered blue carbon ecosystems because of their ability to capture and store a significant amount of carbon (Donato et al., 2011; Lovelock and Duarte, 2019) and provide a range of critical socioeconomic and ecological services to humans (Menéndez et al., 2020). Despite the value of the mangroves, their global spatial extent has decreased by 3363 square kilometers from 2000 to 2016, with 62% of the global losses resulting from land-use changes related to coastal development (Goldberg et al., 2020).

Since mangroves are highly dynamic and their spatial extent and condition changes rapidly (Friess et al., 2019), effective conservation of mangroves requires robust knowledge of their past and current status. Adequate spatial delimitation of ecosystems in the coastal landscape can contribute to better understanding the drivers of its ecological-social dynamic (e.g., ecosystem services; Collins et al., 2010). Monitoring mangrove extent provides baseline data to guide spatial planning decisions and financial investments that mitigate the effects of global environmental change and coastal development (Hsu et al., 2020).

The use of remote-sensing geographic and cartographic tools enables

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long-term and cost-effective mapping and monitoring of land cover and land use at different temporal and spatial scales (Pham et al., 2019). Managers and decision-makers typically use knowledge derived from maps to manage ecosystems (Cvitanovic et al., 2015). Consequently, accurate maps are crucial to guide the conservation and management planning process, including assessing and monitoring evidence-based decision-making (see Congalton and Green, 2019; Rebelo et al., 2009). Reporting uncertainty and errors in the estimation of mangrove extent is critical to assessing the deforestation and degradation processes, changes in carbon stocks, and carbon emissions (Macreadie et al., 2019; Sasmito et al., 2019). This information is also essential to improving land use and environmental policy decisions regarding the conservation of mangroves at different spatial and temporal scales.

Robust mangrove extent data is essential to define successful management and conservation strategies and policies at the national, regional, and global levels (Ruiz-Luna et al., 2008). Several authors have identified the uncertainty of mangrove spatial data and deforestation statistics (e.g., Ruiz-Luna et al., 2008; Friess and Webb, 2011, 2014; Mejía-Rentería et al., 2018). For Mexico, estimates of mangrove losses in the twentieth century ranged from 12% to 60% of the total coverage. This wide disparity in the estimates is due to the different methods used to map mangroves (Ruiz-Luna et al., 2008). In some countries, data discrepancies of mangrove extent have impacted the media and led to the propagation of confounding information to the public (e.g., mangrove spatial extent on the Colombian Pacific coast; Mejía-Rentería et al., 2018). Generating robust mangrove extent data is critical to the policy-making process (Friess and Webb, 2011).

Mexico ranks fourth globally in terms of the highest mangrove extension (Bunting et al., 2018). Thus, to provide official estimates of national mangrove coverage for the five-year intervals since 2005 on a 1:50,000 cartographic scale, the National Commission for the Knowledge and Use of Biodiversity (CONABIO) developed the Mexican Mangrove Monitoring System (MMMS) in 2005 (Acosta-Velázquez and Rodríguez-Zúñiga, 2007). The MMMS is a prestigious and recognized example of interoperability between geographical and mangrove sciences (Latin America Geospatial Forum, 2014), and it currently provides national statistics and reports as well as historical mangrove coverage records and maps. The MMMS provides public access to its databases and applies the same mangrove coverage classification methods for all observation periods to ensure data consistency, compare different periods, and, consequently, estimate reliable deforestation rates across scales.

Surprisingly, the recently published mangrove coverage map for 2020 estimated a total area of 905,086 ha of mangrove coverage at a national scale, which was 129,531 ha (14.3%) more than in 2015 and 48,681 ha (5.4%) more than the 1970–1980 period (Velázquez-Salazar et al., 2021). The gains registered do not necessarily represent "a historical recovery of the mangrove," as CONABIO states in their report (i.e., Velázquez-Salazar et al., 2021). Although these statistics are considered official in Mexico, the accuracy of the mangrove cartography has not been updated as a result of COVID-19 regulations limiting the collection of field validation datasets (Velázquez-Salazar et al., 2021).

Velázquez-Salazar et al. (2021) contend that the expansion of mangrove coverage areas is due to technical improvements of the satellite products and feedback of The National Commission of Natural Protected Areas (CONANP) staff. However, they have not reported the accuracy assessment with aerial photo identification available from free and public CONABIO databases (CONABIO, 2021a). Accuracy assessments of mangrove coverage areas based on photo interpretation are reliable and essential without ground truth validations and measurements (Congalton and Green, 2019).

The improvement of technology is essential to increase the accuracy of mangrove maps. However, the delimitation of mangroves (highly dynamic in the coastal zone) should adhere to consistent criteria to compare mangrove mapping products (Friess and Webb, 2011; Ruiz-Luna et al., 2008).

We hypothesized that these coverage changes could be due to changes in mangrove mapping criteria and definitions. Therefore, the objective of this study was to evaluate the accuracy of changes in the mapping criteria and the definition of mangrove, analyzing mangrove coverage with airborne datasets in one of the largest and most carbon-rich coastal wetlands in the country (i.e., the Sian Ka'an Biosphere Reserve: SKBR). In addition, we discussed the possible sources of discrepancies and implications of these changes at different scales and in different sectors (i.e., government, private, and social) and the possible effects of these changes on the conservation and management of mangroves on a national scale.

2. Material and methods

For the comparative assessment, we downloaded the 2015 and 2020 mangrove coverage national maps and their adjacent coastal areas (i.e., five-kilometer buffer from mangrove map polygons) in Mexico from the CONABIO geoportal (CONABIO, 2016, 2021a, 2021b). These national maps include nine land cover classes (i.e., Anthropic Development, Crop and Animal Husbandry, Other Vegetation Types, Unvegetated, Mangroves, Disturbed Mangroves, Other Wetlands, Water Bodies and Other) (see Valderrama et al., 2014). We then overlaid the 2015 and 2020 maps with each other using conventional geographical information procedures. The result was a map of changes between the 2015 and 2020 periods. Subsequently, the map of changes was validated with a total of 220 randomly generated validation points of airborne data (i.e., vertical/infrared and oblique photos from 2016 MMMS databases; CONABIO, 2021a). The Yucatán Peninsula region was analyzed because it has the greatest mangrove extension in Mexico (Valderrama-Landeros et al., 2020). Finally, each validation point was visually interpreted and compared to the map of changes and the overall accuracy was estimated (Congalton and Green, 2019).

3. Results and discussion

The comparative assessment detected a gain of 129,531 ha of new mangrove coverage. The gains registered represented a change from the Other Wetlands class to the Mangroves class. The map of changes registered 176 airborne validation points that were Other Wetlands and only 44 that were Mangroves (N = 220). The overall accuracy of the map of changes was of 20%, consequently 80% (176 from a total of 220 airborne points) of the new mangrove area gained in the 2020 map from the Other Wetlands class was mapped incorrectly and reclassified to the Mangroves class. These findings are relevant because erroneous estimates can generate misleading interpretations of deforestation rates in the analyzed area (i.e., Yucatán Peninsula Region with the highest economic growth in the country).

The majority of gains were detected in the Yucatán Peninsula region (120,052 ha; 93% of total changes compared with 2015). Specifically, 102,610 ha of the new mangrove areas (79% of total changes) were detected in the SKBR and its surroundings areas.

The gains detected in the SKBR are in zones where herbaceous vegetation, such as *Cladium jamaicense* and *Typha domingensis*, dominates the landscape, and shrubs of *Rhizophora mangle* are found in a dispersed pattern (Fig. 1). This type of vegetation is officially recognized as a "Tular" (INEGI, 2017) and has been reported as "Marshes" (Adame et al., 2013). The MMMS (1970–1980, 2005, 2010, and 2015) included this type of vegetation in the Other Wetlands class, which is defined as "hydrophytic vegetation known as Popal-Tular-Carrizal, flooding grasslands, other hydrophytic or halophytic vegetation types containing dispersed individuals or isolated grooves of mangrove, and coastal saline terrains with sparse vegetation" (Valderrama et al., 2014). After an extensive collaborative effort between academic and public representatives to define the effective criteria for delineating the spatial extent of areas dominated by mangroves, the consensus was reached regarding the definitions of the Mangroves and Other Wetlands classes

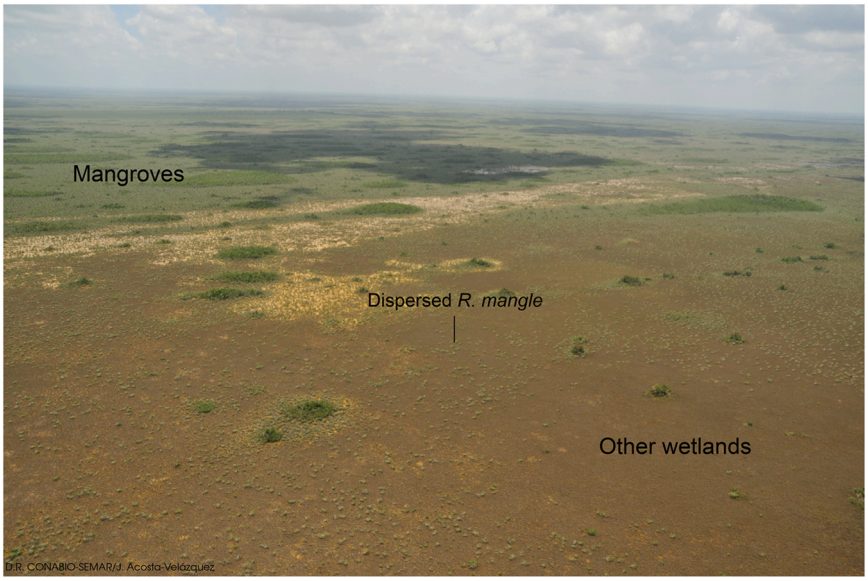

Fig. 1. Aerial photo of Mangroves and Other Wetlands in Sian Ka'an Biosphere Reserve (SKBR).

(Acosta-Velázquez and Rodríguez-Zúñiga, 2007).

Velázquez-Salazar et al. (2021) have explained that the extent of area gains is due to the use of different satellite imagery (SPOT 5 in 2015 vs. Sentinel 2 in 2020) and technical improvements that distinguish mangroves at such sites. However, using our comparative assessment, we found that the majority (93%) of gains in the Mangroves class was explained by the changes in the definitions of the Mangroves and Other Wetlands classes in the 2020 map.

The spatial extent of mangrove coverage areas estimated by the MMMS has been previously validated with an accuracy higher than 87% by regional mangrove experts and by interpreting digital aerial photography (i.e., products 1970–1980, 2005, 2010, and 2015) (Valderrama-Landeros et al., 2020). Differences in the mapping criteria and the definitions of mangroves between assessment years (i.e., 2015–2020) can explain the expansion in mangrove extent.

Even with remote sensing, defining mangrove area spatially can be a challenge at the national, regional, and global scales because mangroves are distributed in complex coastal landscapes interconnected with other coastal habitats (e.g., salt marshes, coral reefs, seagrasses, freshwater wetlands), intertidal mudflats, open water (e.g., coastal lagoons, creeks, rivers, and pools) (Spalding et al., 2010; Friess and Webb, 2014), and other land uses (e.g., coastal development, agriculture, aquaculture). Accurate estimates and trend analyses are critical for the viable implementation of land use and conservation policies for coastal wetlands that exist in constantly fluctuating environments and that experience complex drivers of change (e.g., hydrodynamic forcing, sediment starvation, and sea-level rise) that are not experienced by other ecosystems (Friess and Webb, 2011). Currently, the MMMS has provided uncertain mangrove coverage estimates in 2020, and there is limited ecological support to consider other wetlands as mangroves. Although the presence of *R. mangle* in other wetlands may provide essential ecosystem services (without undermining the potential ecosystem services provided by other wetlands), those wetlands have a different biophysical expression and species composition compared with mangroves (Fig. 1).

Robust spatial data and monitoring information at various temporal, spatial, and governance scales are necessary to develop and implement effective habitat policy for conserving, managing, and restoring valuable mangrove ecosystems (Friess and Webb, 2011; Friess et al., 2016; Mejía-Rentería et al., 2018). Discrepancies between mangrove extent estimates in the MMMS results could have negative implications for implementing effective strategies and policies for the management and conservation of mangroves in Mexico. On a global scale, mangrove extent is used as a metric to achieve international conservation commitment changes in mangrove coverage. Furthermore, mangrove deforestation rates impact the country's international commitments to protect biodiversity, wetlands, and global environmental change. As Mexico ranks fourth globally in the mangrove area, mangrove conservation can contribute to Mexico's Nationally Determined Contributions (updated in 2020 for protecting and restoring blue carbon ecosystems) (SEMARNAT, 2020). A major implication is that MMMS mangrove coverage reports and maps address coastal management issues at the regional and national scales. Thus, the 2020 mangrove map could offer an incorrect or misleading impression to politicians, stakeholders, private sector entities, and members of the public that Mexico has gained at least 16% more mangrove coverage when compared to the 2015 mangrove coverage report. The mangroves of the Pacific coast of Colombia underwent a similar misleading communication of results (Mejía-Rentería et al., 2018).

The misinterpretation of a sham increase of mangrove extent can result in erroneous territorial planning (e.g., blue carbon projects in zones that are other wetlands) and can impact economic activities such as tourism and infrastructure building permits. This discrepancy creates the expectation that public policies governing mangroves are efficient, although these ecosystems are continuously degraded and deforested in Mexico at the local scale (e.g., Dos Bocas Oil Refinery construction in ~40 ha of mangroves) (Ibarzábal and Bonilla, 2020). Discrepancies in the classification criteria used for Other Wetland and Mangrove classes should be addressed in multiple MMMS reports across years to facilitate the precise monitoring and forecasting of mangrove coverage. A cartographic solution is to separate flooded grassland areas with the presence of *R. mangle* from all MMMS products and reclassify them as "other wetlands with the presence of *R. mangle*." Thus, MMMS products would be comparable over time and serve as a tool in the design and implementation of land use and conservation policies of mangroves and other wetlands in Mexico.

The consistency of mangrove extent data is critical to adapt and implement efficient land use and conservation policies for mangrove ecosystems and coastal landscape management at the national, regional, and global levels and to allow for public and private investments of financial and human resources for the monitoring, maintenance, and preservation of the threatened mangrove zones (Ruiz-Luna et al., 2008; Friess and Webb, 2014). Measurable and transparent mangrove conservation targets require an accurate baseline of historical data (Friess and Webb, 2011), particularly for payments for ecosystem services

schemes. In the face of the climate and global change crisis, robust data on mangrove extent is of particular interest to assess contributions to the global, national, and regional coastal carbon cycle.

Finally, the definition and spatial delimitation of ecosystems and temporal scales must be consistent. The mapping criteria must be detailed for comparative purposes. Analyses of the maps and official statistics (e.g., deforestation rates, landscape metrics) are used by stakeholders and policymakers to design and implement policies and other tools related to the use, management, and preservation of ecosystems.

4. Conclusions

This work highlights the presence of discrepancies in official mangrove coverage estimates across years associated with mangrove mapping and definition criteria. This analysis demonstrates that the significant increase in mangrove coverage in Mexico is due to changes in mangrove mapping criteria and mangrove definitions. The differences between mangrove coverage in Mexico (i.e., 2015 vs. 2020) are not actual gains but are rather the result of changes in the mangrove mapping criteria and the definition of "mangrove." We propose to correct estimates across periods and precisely detect trends in the mangrove-dominated areas (i.e., mangrove community). We recommend that mangrove coverage estimates be peer-reviewed before being published to avoid misleading interpretations by multiple users of information. The official cartographic products at the national scale must be consistent to avoid discrepancies and ensure different sectors' correct use and perception.

The official statistics must be validated and consistent to formulate environmentally efficient public policies. Moreover, these public policies must consider each type of wetland's specific ecological dynamic for adequate management. The spatial distribution of mangrove wetland ecosystem services is related to each system's intrinsic biophysical attributes (e.g., biomass, structure; *sensu* Ochoa-Gómez et al., 2019), which has ecological and economic implications. Inclusively, the legal protection could be self-defeating to adequately manage herbaceous ecosystems (e.g., marshes) with carbon dynamics different from woody plants such as mangroves. Capital investment for mangrove restoration programs in areas where other wetlands dominate has important implications for different scales/sectors. Herbaceous plants (e.g., marshes) provide specific ecosystem services that are not replaceable by mangroves and vice-versa. Sites such as the SKBR could be susceptible to capital investment programs for mangrove restoration programs that will modify/alter the ecosystem services of other wetlands.

Finally, it is essential to consider the impact of changing the mangrove mapping criteria on carbon budget/dynamics, restoration interventions, ecosystem services, deforestation rates, and socioeconomic implications.

Data Availability

Supplementary data available.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.landusepol.2023.106651.

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